Questionnaire Metrics & Data Mapping Guideline

This guideline maps every questionnaire field to its role in the AI-driven clustering pipeline and to the dashboard fields it feeds. All question references use the current questionnaire numbering (Q1–Q27).

1. AI Data-Driven Cluster Discovery Framework

The system discards chronological year-of-experience (Q5) grouping boundaries. An unsupervised K-means (or density-based, e.g. DBSCAN) clustering algorithm processes multi-dimensional feature vectors to categorise agents based on behavioural footprints, tool workarounds, and perceived digital utility. The number of clusters it discovers, K, is chosen dynamically per run — via a data-driven criterion such as Silhouette-score optimisation — and is not fixed to any specific count. Demographic variables such as Age (Q1), Position (Q4), and Tenure (Q5) are treated strictly as descriptive characteristics to examine within each discovered cluster, never as hardcoded segregation boundaries.

Dashboard Analysis Logic

Each agent's response maps to a four-dimensional vector R = [SUP, WP, AR, PR]. The clustering model calculates the distance of R from each discovered cluster's centroid to assign agents to clusters, and analysts can cross-tabulate discovered clusters against tenure/age/position to see whether, for example, high-tenure agents actually group into manual-workaround behaviour or into advanced-optimiser behaviour — without that outcome being assumed in advance.

2. AI Classification Vector Definitions

Each formula resolves to a single, unambiguous question or field set — no formula spans unrelated question types, and every referenced field actually exists at the location the formula names.

System Utility Perception (SUP)

Formula: Mean score of Q8, Q10, and all sub-items inside the Q11 matrix (Digital Tool Usage, System/Portal Speed & Reliability, Mobile App Experience).

Workaround Propensity (WP)

Formula: Aggregated indexing of multi-select counts in Q13 (reasons for bypassing the company system), Q14 (personal-app/paper-calendar selections and stated reasons), and Q23 (Personal Tools / Workarounds list).

Abandonment Rate (AR)

Formula: Normalized count of system dropouts and pain-point mentions extracted via keyword/semantic tracking from Q27(c) — the pain-point field — across all six journey-map tasks.

Peer Reliance (PR)

Formula: Frequency index of “Senior colleague” or “Administrative Assistant / Peer support” selections made under Q27(b) across all task actions, cross-validated against the “Administrative Assistant / Peer assistance for data logging” option in Q23.

3. Dashboard Field Definitions

Agent Name

- Input Type: Input Box
- Logic: Manually entered on the dashboard (not an item from the questionnaire).

Position / Title

- Corresponds to: Q4
- Formula: The mode (most-selected) option in Q4, calculated dynamically across rows matching the corresponding discovered-cluster assignment column (_clusterId).

Agent Sales Journey

- Corresponds to: Q26
- Logic: Categorised distribution percentage of chosen operational focus (Sales vs Organisational), segmented by discovered cluster.

Tag 1, Tag 2, Tag 3 — Self-Image Style Tags

- Corresponds to: Q17, Q18, and Q19, respectively
- Formula: Top-ranked frequency distribution of tag selections, calculated dynamically for each discovered cluster.

Device / System Familiarity

- Corresponds to: Q8
- Formula: Average familiarity percentage calculated for each discovered cluster.

Daily Tasks Allocation

- Corresponds to: Q20
- Logic: Calculations are mapped via the following long-format aggregation:
  - Step 1: Reshape to long format: one respondent | one activity | its estimated %.
  - Step 2: Map raw selection bracket midpoints to standardise percentage metrics.
  - Step 3: Compute Mention Rate (% of cluster mentioning task) and Average Allocation (mean percentage within cluster).
  - Step 4: Sort and rank primary operations per discovered cluster.

Core Barriers / Contradictions in Daily Tasks

- Corresponds to: Q21 (dedicated open-text field)
- Data Collection: Automated word cloud and text semantic clustering of the Q21 open-entry text, grouped by discovered cluster.

Agent Journey Performance & Emojis

- Corresponds to: Q27, Tasks I–VI
- Formula: Mean rating scores calculated across all matching cluster rows.
  Mapped emojis dynamically transform based on the 1–5 score boundaries (
  1 = 😫, 2 = 😮‍💨, 3 = 😐, 4 = 🙂, 5 = 🤩).
- Implementation note: the rating item's letter varies by task — it is “D” for the five tasks with 3 actions (I, III, IV, V, VI) and “C” for Task II, which has only 2 actions. Parsers should key off the task's last item, not a fixed letter.

Cluster Count & Labels (dashboard-level)

- Corresponds to: output of the clustering job, not a fixed questionnaire field.
- Logic: the dashboard should render one card/colour per cluster actually returned by the current clustering run, with centroid values and labels computed live rather than read from a fixed lookup table.